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Ex.No. 12: NB-IoT vs LTE-M Throughput Simulation Date

AIM: To simulate and compare the performance of **NB-IoT and LTE-M communication technologies** in terms of **Throughput (Mbps), Latency (ms), and Packet Delivery Ratio (%)** using MATLAB.

Objective:

1. To evaluate and compare the **Throughput (Mbps)** performance of NB-IoT and LTE-M under varying network conditions.
2. To analyze the **Latency (ms)** characteristics of NB-IoT and LTE-M and study their impact on real-time communication.
3. To measure and compare the **Packet Delivery Ratio (%)** of NB-IoT and LTE-M systems for reliability assessment.

Procedure

1. Define system parameters for NB-IoT and LTE-M such as bandwidth, transmission power, and modulation efficiency.
2. Generate network conditions by varying Signal-to-Noise Ratio (SNR) or channel quality values.
3. Compute **Throughput (Mbps)** for both NB-IoT and LTE-M using their respective data rate models.
4. Calculate **Latency (ms)** based on transmission time, scheduling delay, and network load.
5. Determine **Packet Delivery Ratio (PDR %)** by comparing successfully received packets with transmitted packets.
6. Simulate multiple iterations to observe performance under different channel conditions.
7. Plot comparison graphs for Throughput, Latency, and PDR between NB-IoT and LTE M.
8. Analyze results to identify which technology performs better under different IoT Scenarios

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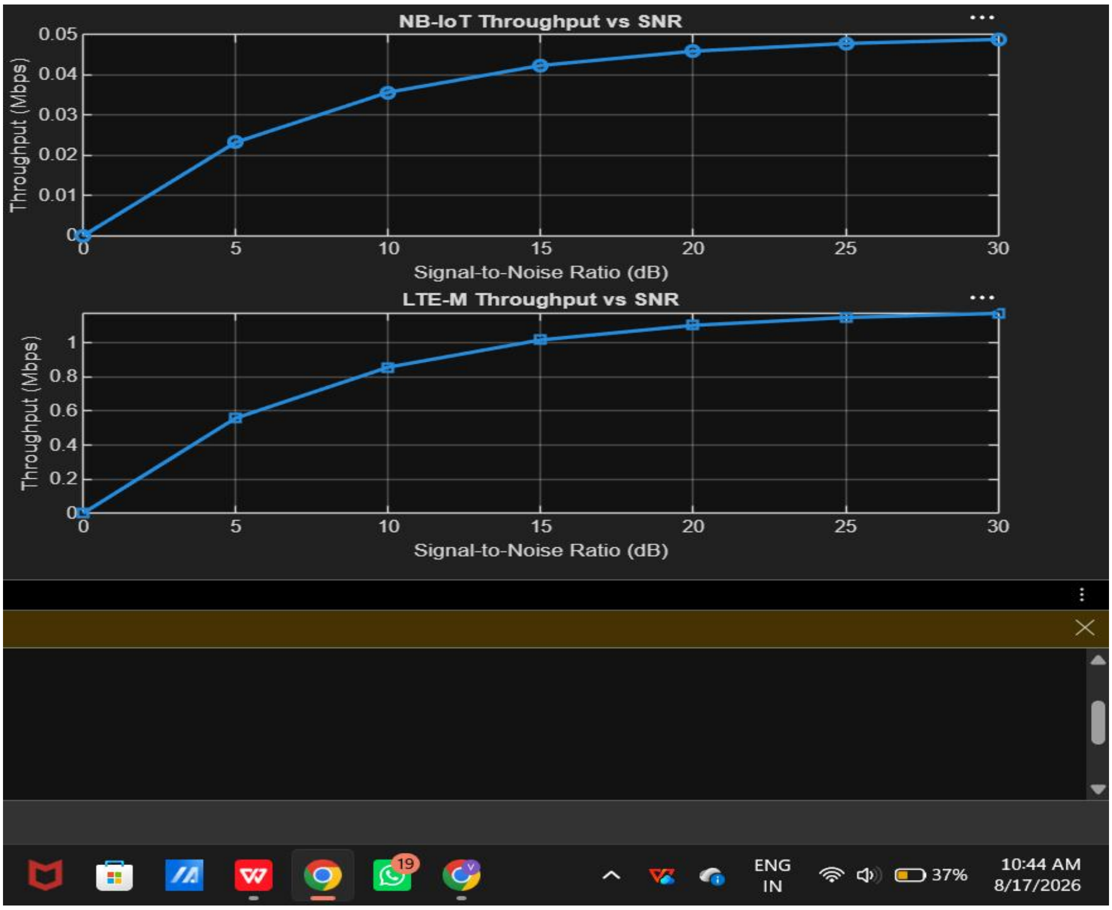
Objective 1 Matlab Code and Output:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Throughput (Mbps)
NB_IoT = 0.05*(1-exp(-SNR/8));
LTE_M = 1.2*(1-exp(-SNR/8));disp('SNR (dB) NB-IoT(Mbps) LTE-M(Mbps)')
disp([SNR' NB_IoT' LTE_M'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
```

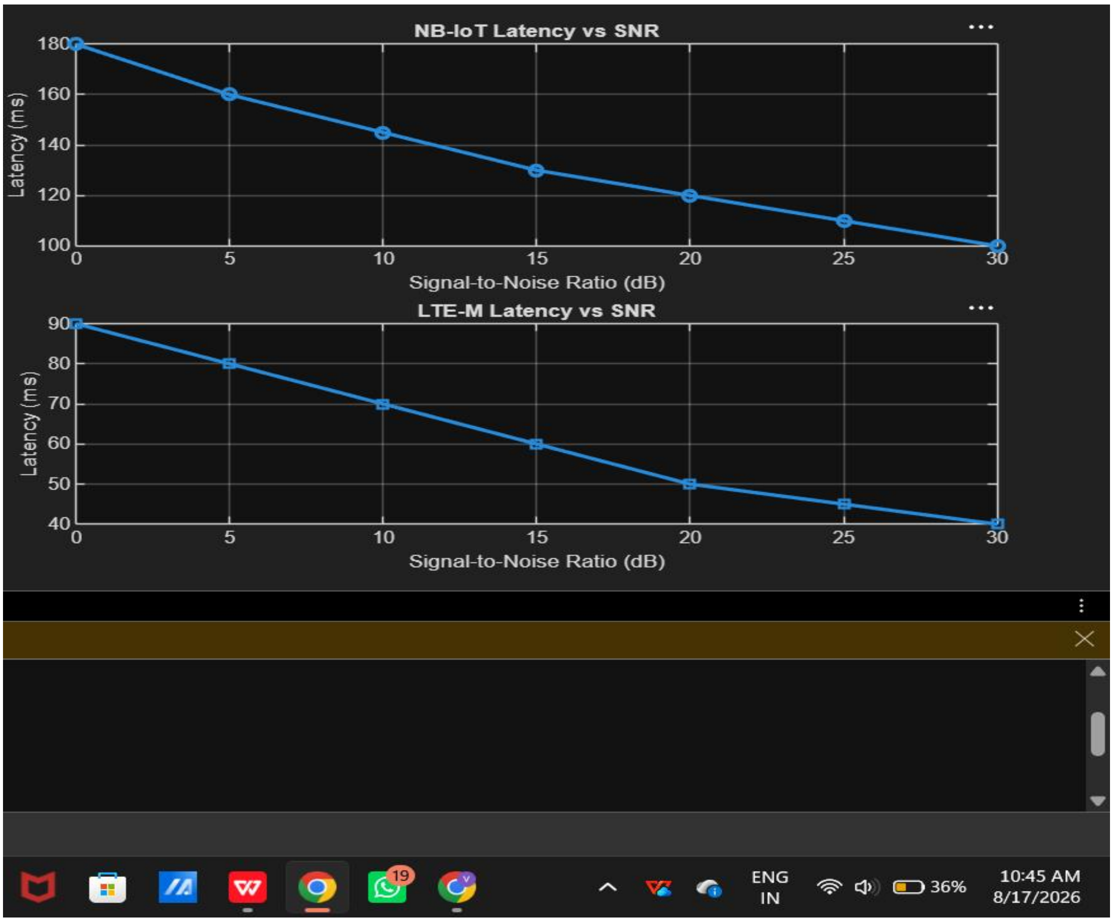


The throughput of both NB-IoT and LTE-M increases as the Signal-to-Noise Ratio (SNR) improves. LTE-M achieves significantly higher throughput than NB-IoT, making it more suitable for applications requiring higher data rates.

Objective 2 Matlab Code and Output:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Latency (ms)
NB_IoT = [180 160 145 130 120 110 100];
LTE_M = [90 80 70 60 50 45 40];
disp('SNR(dB) NB-IoT Latency(ms) LTE-M Latency(ms)')
disp([SNR' NB_IoT' LTE_M'])
figure
% -------- Graph 1 --------
```

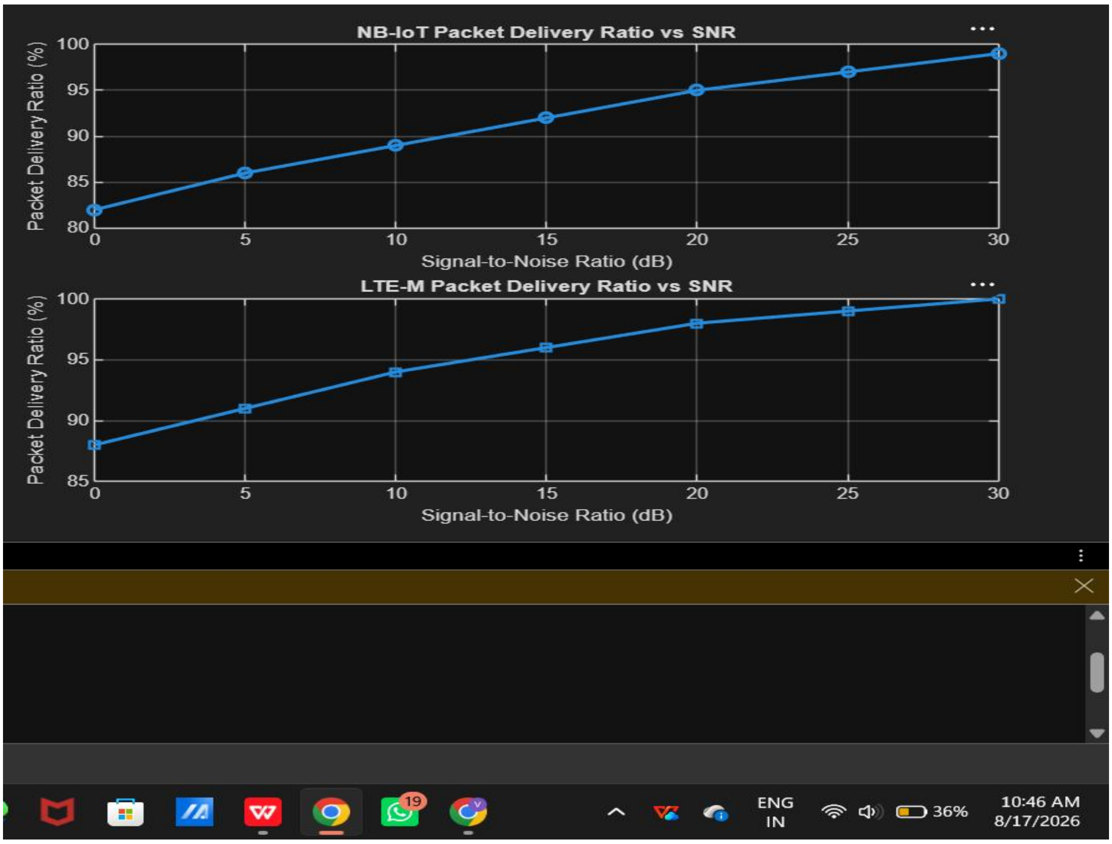
```
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on
```



The latency of both NB-IoT and LTE-M decreases as the SNR increases due to improved channel quality. LTE-M exhibits lower latency than NB-IoT, making it more suitable for delay sensitive IoT applications such as real-time monitoring and asset tracking.

Objective 3 Matlab Code and Output:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Packet Delivery Ratio (%)
NB_IoT = [82 86 89 92 95 97 99];
LTE_M = [88 91 94 96 98 99 100];
disp('SNR(dB) NB-IoT PDR(%) LTE-M PDR(%)')
disp([SNR' NB_IoT' LTE_M'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
% -------- Graph 2 --------subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
```



The Packet Delivery Ratio (PDR) of both NB-IoT and LTE-M increases as the Signal-to-Noise Ratio (SNR) improves. LTE-M achieves a slightly higher PDR than NB-IoT, indicating more reliable packet transmission under the same channel conditions.

Result:

1. The throughput performance of both NB-IoT and LTE-M was successfully simulated, and LTE-M achieved higher throughput than NB-IoT under the same network conditions.

2. The latency analysis showed that LTE-M provides lower communication delay compared to NB-IoT, making it more suitable for real-time IoT applications.

3. The Packet Delivery Ratio (PDR) of both technologies improved with increasing SNR, with LTE-M demonstrating slightly higher reliability than NB-IoT during data transmission.